Ischemic Heart Disease & Acute Coronary Syndromes

Page 1: Coronary Atherosclerosis & Plaque Vulnerability

Pathogenesis of Thin-Cap Fibroatheroma (TCFA)

High-Yield Clinical Concepts & Board Review Points:

- **Endothelial injury leads to LDL accumulation, oxidation, and macrophage foam-cell formation.**
- **Fibrous cap thinning is mediated by macrophage matrix metalloproteinases (MMP-1, MMP-9, MMP-13).**
- **Plaque rupture triggers immediate exposure of tissue factor and subendothelial collagen to circulating blood.**
- **Platelet adhesion occurs via glycoprotein Ib-IX to von Willebrand factor (vWF).**

CLINICAL PEARL & EXAM TRAP:

Atherosclerotic plaques with core necrotic areas >25% and caps <65 microns are at highest risk of acute rupture.

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Page 2: STEMI vs NSTEMI Biomarkers & ECG Localization

Troponin Kinetics and Coronary Vessel Territories

High-Yield Clinical Concepts & Board Review Points:

- **ST elevations in II, III, aVF localize acute transmural ischemia to the inferior wall (RCA in 85%).**
- **ST elevations in V1-V4 indicate anterior wall infarction via the Left Anterior Descending (LAD) artery.**
- **Cardiac troponin I and T rise within 2-4 hours, peak at 24 hours, and stay elevated for up to 10-14 days.**
- **Creatine kinase-MB (CK-MB) returns to baseline within 48-72 hours, making it ideal for detecting reinfarction.**

CLINICAL PEARL & EXAM TRAP:

In inferior STEMI, always obtain right-sided ECG leads (V4R) to rule out Right Ventricular Infarction before administering

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Page 3: Medical Management & Revascularization Timing

Antiplatelet, Anticoagulant, and PCI Protocols

High-Yield Clinical Concepts & Board Review Points:

- **Door-to-balloon time goal for primary PCI is <90 minutes at PCI-capable centers, or <120 minutes if transferred.**
- **Dual antiplatelet therapy (DAPT): Aspirin plus a P2Y12 inhibitor (Ticagrelor or Prasugrel preferred over Clopidogre**
- **Beta-blockers reduce myocardial oxygen demand and ventricular arrhythmia risk within the first 24 hours.**
- **High-intensity statin therapy (Atorvastatin 80 mg or Rosuvastatin 40 mg) must be initiated regardless of baseline L**

CLINICAL PEARL & EXAM TRAP:

Contraindications to early beta-blocker therapy include signs of acute heart failure, low output state, PR interval >0.24s, o

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Page 4: Mechanical Complications of Myocardial Infarction

Day 3 to 7 Post-Infarction Structural Failures

High-Yield Clinical Concepts & Board Review Points:

- **Papillary muscle rupture (days 3-5): Causes acute severe mitral regurgitation, holosystolic murmur, and pulmonary**
- **Interventricular septum rupture (days 3-5): Presents with harsh holosystolic murmur with thrill at left sternal borde**
- **Free wall rupture (days 5-14): Leads to sudden hemopericardium, cardiac tamponade, electromechanical dissociat**
- **Ventricular pseudoaneurysm: Contained by pericardium/thrombus; requires emergent surgical repair due to high r**

CLINICAL PEARL & EXAM TRAP:

Posteromedial papillary muscle has a single blood supply (posterior descending artery from RCA), making it far more vul